Question ID: 9fa781f8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Medium

Question

The ratio x to y is equivalent to the ratio 9 to 5. If the value of x is 162, what is the value of y?

Correct Answer: 90

Rationale

The correct answer is 90. It's given that the ratio of x to y is equivalent to the ratio 9 to 5. It follows that $\frac{x}{y}=\frac{9}{5}$. Multiplying each side of this equation by $5y$ yields $\frac{(5y)x}{y}=\frac{9(5y)}{5}$, or $5x=9y$. Dividing each side of this equation by 9 yields $\frac{5x}{9}=y$. Substituting 162 for x in this equation yields $\frac{5(162)}{9}=y$, which is equivalent to $\frac{810}{9}=y$, or $90=y$. Therefore, if the value of x is 162, the value of y is 90.